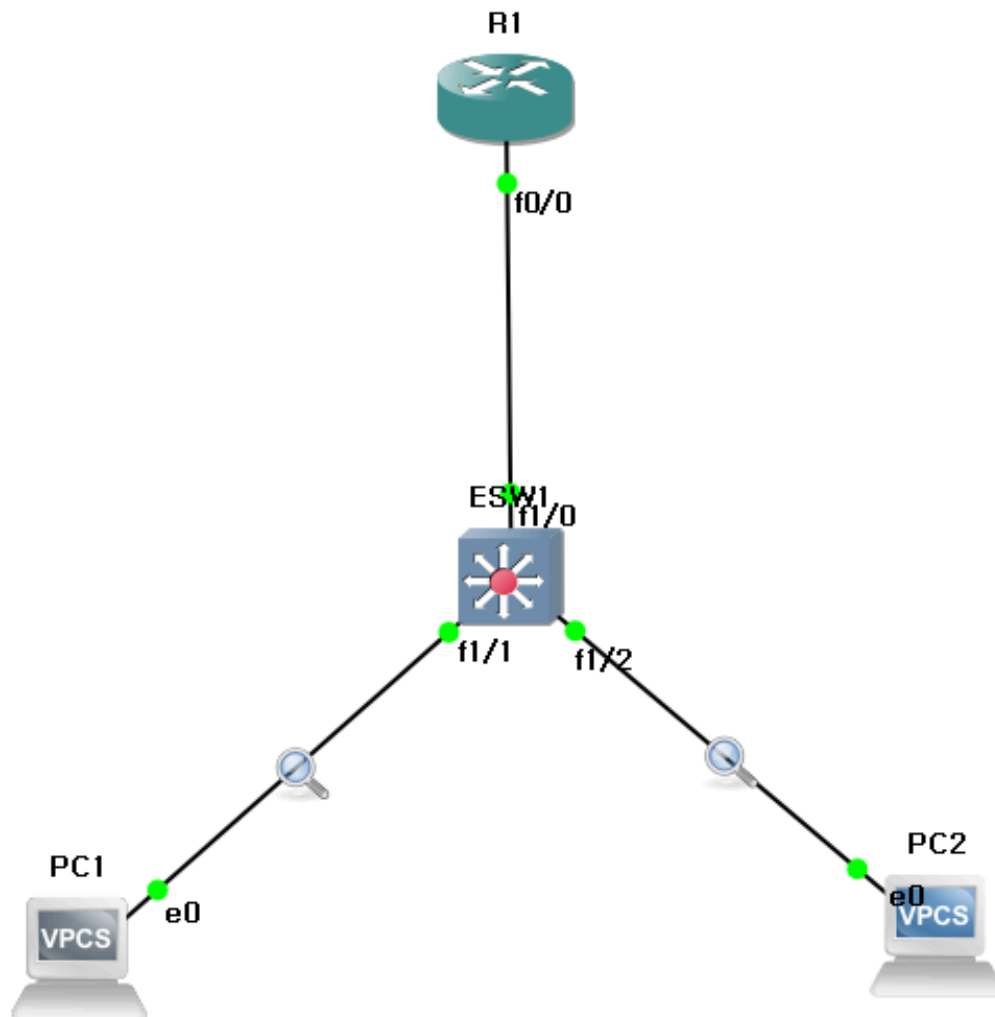


[250902] DHCP 보고서

GNS3에서의 패킷 연결



라우터를 DHCP서버로 설정

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip add 192.168.10.2 255.255.255.0
R1(config-if)#no sh
R1(config-if)#
*Sep  2 17:30:39.371: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
R1(config-if)#
*Sep  2 17:30:39.371: %ENTITY_ALARM-6-INFO: CLEAR INFO Fa0/0 Physical Port Admin
istrative State Down
*Sep  2 17:30:40.371: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config-if)#end
R1#
*Sep  2 17:30:47.927: %SYS-5-CONFIG_I: Configured from console by console
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#inf f0/0
      ^
% Invalid input detected at '^' marker.

R1(config)#ip dhcp pool KONG
R1(dhcp-config)#network 192.168.10.0 255.255.255.0
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#default-router 192.168.10.1
R1(dhcp-config)#ip dhcp excluded-address 192.168.10.1 192.168.10.9
R1(dhcp-config)#exit
R1#
*Sep  2 17:32:37.351: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/          Lease expiration    Type
                   Hardware address/
                   User name
```

- 글로벌 설정모드 (장비 전체에 영향을 주는 설정)

```
R1#conf t > R1(config)#
```

- 해당 인터페이스로 이동 → 글로벌 설정모드에서 대상 인터페이스의 이름을 입력

```
R1(config)#int f0/0 > Router(config-if)#
```

- ip 설정

```
Router(config-if)#ip add 192.168.10.1 255.255.255.0
```

```
Router(config-if)#no sh
```

```
Router(config-if)#end
```

<DHCP 설정>

- dhcp pool을 생성

```
R1(config)#ip dhcp pool KONG
```

- NI 등록

```
R1(dhcp-config)#network 192.168.10.0 255.255.255.0
```

- DNS 등록

```
R1(dhcp-config)#dns-server 8.8.8.8
```

- GW 등록

```
R1(dhcp-config)#default-router 192.168.10.1
```

- 제외 IP 등록

```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9
```

----- : 제외 IP 192.168.10.1 ~

192.168.10.9

- 현재 상태

```
R1#sh ip dhcp binding
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
------------	--------------------------------	------------------	------

PC1 DHCP 서버 요청

```
PC1> ip dhcp
DORA IP 192.168.10.10/24 GW 192.168.10.1
```

```
PC1> ip dhcp
```

와이어샷크 패킷 캡처

1	0.000000	Private_66:68:01	Broadcast	ARP	64 Gratuitous ARP for 192.168.10.11 (Request)
2	1.000269	Private_66:68:01	Broadcast	ARP	64 Gratuitous ARP for 192.168.10.11 (Request)
3	2.013670	Private_66:68:01	Broadcast	ARP	64 Gratuitous ARP for 192.168.10.11 (Request)
4	2.852257	c4:02:02:1c:f1:02	Spanning-tree-(for-...	STP	60 Conf. Root = 32768/0/c4:02:02:1c:00:00 Cost =
5	8.296849	c4:02:02:1c:f1:02	Spanning-tree-(for-...	STP	60 Conf. Root = 32768/0/c4:02:02:1c:00:00 Cost =
6	9.844380	0.0.0.0	255.255.255.255	DHCP	406 DHCP Discover - Transaction ID 0x4929c41e
7	9.861509	192.168.10.2	192.168.10.11	DHCP	342 DHCP Offer - Transaction ID 0x4929c41e
8	10.858928	0.0.0.0	255.255.255.255	DHCP	406 DHCP Request - Transaction ID 0x4929c41e
9	10.874724	192.168.10.2	192.168.10.11	DHCP	342 DHCP ACK - Transaction ID 0x4929c41e
10	11.873270	Private_66:68:01	Broadcast	ARP	64 Gratuitous ARP for 192.168.10.11 (Request)
11	12.874171	Private_66:68:01	Broadcast	ARP	64 Gratuitous ARP for 192.168.10.11 (Request)
12	13.611203	c4:02:02:1c:f1:02	Spanning-tree-(for-...	STP	60 Conf. Root = 32768/0/c4:02:02:1c:00:00 Cost =
13	13.874499	Private_66:68:01	Broadcast	ARP	64 Gratuitous ARP for 192.168.10.11 (Request)